AP Precalculus

Unit 3: Trigonometric and Polar Functions

Lesson 3.10 Trigonometric Equations and Inequalities

Name: ____________________________ Date: ________________ Period: ________

Learning Targets — I Can...

1. I can solve a trigonometric equation by isolating the trig function, applying an inverse trig function to find the principal value, and using periodicity to write the general solution. *(LO 3.10.A)*
2. I can explain why trigonometric equations have infinitely many solutions when the domain is unrestricted, and identify both solution families. *(LO 3.10.A, EK 3.10.A.2)*
3. I can apply a domain restriction implied by a contextual scenario to select only the solutions that are valid in that context. *(LO 3.10.A, EK 3.10.A.3)*

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral review: inverse trig values, unit circle, isolating trig functions	________
2	Guided Notes	General solutions, inverse trig procedure, contextual domain, inequalities	________
3	Group Activity	Solve trig equations and inequalities (tiered R/A/E)	________
4	Exit Ticket	Solve $\sin\theta = -\frac{\sqrt{3}}{2}$; explain infinitely many solutions	________
5	Homework	Mixed: general solutions, context, AP MC, quadratic substitution extension	________
		Total	________